

Methodology: Share of Wallet Estimation

Team **The Independent Variable** — Daniel Mataranyika, Herton Mabongue Standard Bank × Data School Hackathon 2026

1. The question, and why it is hard

Syn Bank can see every transaction that flows through Syn Bank. It cannot see the transactions that flow through its competitors. So the commercially useful number — *what fraction of this client's banking activity do we hold?* — cannot be read off the internal data at all. It has to be estimated from outside.

This is the trap the model is built to avoid, and it is worth being explicit because our own first version fell into it.

1.1 The circularity we found and removed

Our initial implementation sized each client's total wallet as `internal_flow × benchmark_ratio`, then estimated the captured amount as `total_wallet × capture_ratio`, where `capture_ratio` was a bounded heuristic over transaction counts and channel counts. Substituting one into the other:

```
share = captured / wallet
      = (wallet × capture_ratio) / wallet
      = capture_ratio
```

The wallet term cancels. Share of wallet was mathematically identical to the assumed capture rate and could not depend on the data at all. Empirically, across twenty clients whose estimated wallets differed by a factor of **301×**, the reported share ran from 18.2% to 20.4% — a spread of two percentage points, with a standard deviation of 0.006.

A reviewer who tested only that shares fall between 0 and 1 would never catch this.

`tests/test_wallet_engine.py::test_share_varies_across_clients` is the regression guard we added so it cannot come back.

2. The two-sided model

The fix is to compute the two sides of the ratio from **independent sources**.

SIDE	QUESTION	SOURCE
Addressable flow	How much banking activity <i>should</i> this client generate?	Published annual financial statements (external)
Captured flow	How much did Syn Bank actually see?	Internal transaction datasets (internal)

```
share = captured_flow / addressable_flow
gap   = addressable_flow - captured_flow
revenue opportunity = gap × pillar fee margin
```

Both sides are annual ZAR **flows**, which makes the ratio meaningful: an estimated quantity divided by a measured one, not an assumption restated.

2.1 Top-down: addressable flow

For each client we take the financial-statement figure that drives each product pillar, restrict it to the portion addressable by a South African bank, and apply a sector flow multiple.

PILLAR	FINANCIAL-STATEMENT SIGNAL	RATIONALE
Transactional	Revenue	Collections, supplier payments, payroll and treasury sweeps all cross the bank account. Both sides of the working-capital cycle pass through, so annual flow exceeds annual revenue.
FX / Cross-border	Revenue × foreign revenue share	Exporters repatriate receipts, importers settle abroad, and intercompany treasury adds a further leg.
Trade Finance	Cost of sales + inventory	Documentary instruments (LCs, guarantees, collections) are raised against the physical goods cycle. Only a minority of trade settles under an instrument, so multiples sit well below 1.

```
addressable_pillar = signal × sa_attribution_share × sector_flow_multiple
```

All multiples live in `config/benchmarks.yaml` with low/base/high bands.

2.2 The SA-attribution adjustment

Several clients are global groups. BHP's worldwide revenue is not addressable by a South African bank. We therefore scale each client's signal by an `sa_attribution_share` — our estimate of the portion of activity that plausibly touches a South African banking relationship.

This is the single most judgmental input in the model. It ranges from 0.02 (NEPI Rockcastle, Shaftesbury Capital — property portfolios held almost entirely outside South Africa) to 0.98 (Clicks Group — effectively a domestic-only retailer). It is held in one column of `data/external/company_financials.csv` so a reviewer can challenge any single value without touching the model.

2.3 Bottom-up: captured flow

Captured flow is a straight measurement over the **most recent 12 months** of the internal data, so it is directly comparable to an annual financial-statement figure:

- Transactional: sum of `amount_zar` in `transactional_banking.csv`
- FX: sum of `value_zar` in `cross_border_payments.csv`
- Trade finance: sum of `value_zar` in `trade_finance.csv`

No assumption enters this side. That is the property that makes the sensitivity analysis in §4 meaningful.

3. Opportunity ranking

Ranking on gap size alone sends bankers at the biggest client every time. The composite score in `config/scoring_weights.yaml` blends commercial upside with whether the business is realistically winnable:

WEIGHT	FEATURE	REASONING
50%	Gap size	Commercial upside comes first.

WEIGHT	FEATURE	REASONING
20%	Ease of conversion	An existing multi-product relationship converts more cheaply than a cold pillar.
15%	Relationship depth	Number of active product pillars.
15%	Urgency / freshness	90-day flow acceleration suggests the client is in motion now.

4. Sensitivity to assumptions

Every flow multiple was shifted together and the portfolio re-estimated. The captured side is invariant by construction — it is measured — so only the estimate moves.

MULTIPLIER SHIFT	ADDRESSABLE (RBN)	CAPTURED (RBN)	PORTFOLIO SHARE	REVENUE OPPORTUNITY (RBN/YR)
-25%	3 221	191.6	5.95%	7.12
-10%	3 865	191.6	4.96%	8.64
0%	4 295	191.6	4.46%	9.65
+10%	4 724	191.6	4.06%	10.66
+25%	5 369	191.6	3.57%	12.18

Reading: if every multiple is wrong by a quarter in the same direction, portfolio share still lands between 3.6% and 6.0%. The headline conclusion — that Syn Bank holds a low single-digit share and the growth runway is large — is robust to the assumption set. Client *rankings* are more stable still, because a uniform shift affects all clients in the same direction.

Reproduce with `models.wallet_engine.sensitivity_analysis`.

5. Data sources

Internal (synthetic, supplied): `transactional_banking.csv` (2.80m rows), `cross_border_payments.csv` (241k rows), `trade_finance.csv` (20.3k rows), all covering 2023-07-01 to 2026-06-30 across 20 client entities.

External (public): FY2025 annual results and financial statements for each of the 20 listed entities. Every figure is cited per client in `data/external/company_financials.csv` (`source` column). Currency conversion uses the 2025 average USD/ZAR of 17.8839; EUR and GBP are cross-rates derived from that anchor and are flagged as approximate.

5.1 Treatment of the synthetic-data rules

The brief requires teams to "supplement the supplied internal data with publicly available external sources", naming annual financial reports and investor relations pages, and requires that external sources be cited. It separately prohibits linking the fictional data to any real bank, client or transaction.

We hold both. Public financials are used **only to size an addressable flow benchmark**. We make no claim that any synthetic transaction corresponds to a real event, and Syn Bank is treated throughout as the fictional institution it is. Where the datasets carry the name of a listed entity, we use that entity's published financials as a scale reference and nothing more.

6. Assumptions and limitations

1. **Portfolio size.** The brief describes 50 JSE-listed clients; the supplied data contains 20 (E01–E20). We modelled what was supplied and did not extrapolate.
2. **Flow multiples are benchmarks, not observations.** They are informed by SARB payment-system and banking-fee context but are not client-specific. §4 quantifies the consequence.
3. **SA attribution is judgmental.** See §2.2. It is the input we would most want to replace with disclosed segment data given more time.
4. **Revenue basis differs by industry.** For insurers we use IFRS 17 insurance revenue (Sanlam R292.0bn, OUTsurance R36.0bn); for the property companies, net operating or gross rental income; for Glencore, note that its reported revenue is marketing turnover and is not comparable to a producer's, which is why its SA-attribution share is set very low. Each client's basis is recorded in the `revenue_basis` column. Every figure is now `reported` or `derived` — no estimated revenues remain.
5. **Imputed cost of sales and inventory.** Where not separately disclosed, these are imputed from sector gross margins. Imputation is flagged per client (`cogs_imputed` , `inventory_imputed`) and lowers confidence.
6. **Gaps are attributed to competitors.** We assume Syn Bank's internal records are complete, so unobserved flow sits with another bank. In reality some of it is self-funded, netted internally, or does not exist.
7. **Share above 100% is a signal, not a cap.** If captured exceeds addressable, the top-down estimate under-sized that client. We flag (`share_exceeds_estimate`) rather than silently clipping. No client currently trips it.
8. **Lending and DCM are out of scope.** The supplied data covers three pillars; debt is captured in the external dataset but not yet modelled as a wallet.
9. **Nominal ZAR.** No inflation adjustment within the hackathon window.